

# TAU-MULTILINGUAL: BENCHMARKING VOICE AGENTS ACROSS LANGUAGES

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## ABSTRACT

English-only benchmarks expose only a narrow slice of voice-agent behavior. We introduce  $\tau$ -Multilingual, extending  $\tau$ -Voice to Spanish, Brazilian Portuguese, Hindi, Korean, and Mandarin with native-speaker review and evaluation of generated language and spoken output. Across 4,500 full-duplex calls and five voice configurations, Spanish, Portuguese, and Hindi remain within 3.2 task-completion points of English, but Korean and Mandarin fall by 14.7 and 8.4 points. The failure modes also vary: Korean systems miss more responses, Mandarin systems interrupt more often, and both struggle with tools and entities. Grok leads task completion but scores lowest on generation quality, motivating separate task, interaction, and generation reporting. We release language packs, validated judges, and tools for community-built multilingual voice-agent evaluation.

**Index Terms**— multilingual voice agents, spoken dialogue evaluation, task-oriented dialogue, full-duplex interaction, user simulation

## 1. INTRODUCTION

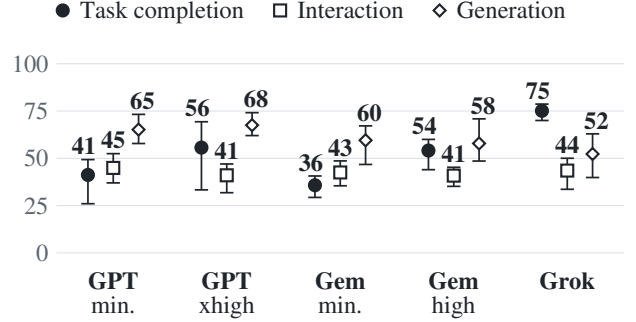
Voice agents must serve a multilingual world: English is spoken by only 19% of the world’s population [1], yet leading end-to-end voice-agent benchmarks evaluate English alone [2, 3, 4]. Multilingual evaluation must therefore capture task completion, conversational behavior, linguistic naturalness, and speech fidelity.

We introduce  $\tau$ -Multilingual, extending  $\tau$ -Voice [2] with three contributions. First, *LanguageFactory* provides a reviewed workflow for native-speaker-authored simulators, localizing caller personas, speech rules, and identities while preserving the underlying tasks. Second, we add human-validated evaluation of linguistic naturalness and speech fidelity alongside task completion and interaction measures. Third, we evaluate five voice configurations and two text systems across six languages and three domains.

We release the codebase [5] so the community can build callers reflecting local linguistic and cultural norms, rather than merely translated English personas, alongside language-specific audits of English-centric agent behavior.

## 2. RELATED WORK

$\tau$ -Voice brings the database-grounded tasks of  $\tau$ -bench and  $\tau^2$ -Bench [13, 14] to full-duplex speech, evaluating task completion and conversational dynamics but not generation quality [2]. It, EVA-Bench, and FDB-v3 are English-only [3, 4]. M3-DuplexBench covers English and Japanese turn-taking without



**Fig. 1.** Fixed-system means with worst-to-best language ranges. Task completion and Interaction average six languages; Generation averages the five localized languages.

**Table 1.** Related benchmarks. A check mark denotes direct evaluation and a tilde partial coverage. Spk.: spoken; FD: full-duplex runtime; Mult.: multilingual; IQ: interaction-quality evaluation; Lang. quality: naturalness and language use.

| Work                                    | Spk. | FD | Mult. | Task | IQ | Lang. quality |
|-----------------------------------------|------|----|-------|------|----|---------------|
| <i>Voice-agent benchmarks</i>           |      |    |       |      |    |               |
| $\tau$ -Voice [2]                       | ✓    | ✓  | —     | ✓    | ✓  | —             |
| FDB-v3 [4]                              | ✓    | ✓  | —     | ✓    | ✓  | —             |
| EVA-Bench [3]                           | ✓    | ~  | —     | ✓    | ✓  | ~             |
| M3-DuplexBench [6]                      | ✓    | ✓  | ✓     | —    | ✓  | —             |
| <i>Multilingual speech and dialogue</i> |      |    |       |      |    |               |
| VoiceAgentBench [7]                     | ✓    | —  | ✓     | ~    | —  | —             |
| Speech-MASSIVE [8]                      | ✓    | —  | ✓     | —    | —  | —             |
| X-RiSAWOZ [9]                           | —    | —  | ✓     | —    | —  | —             |
| GlobalWoZ [10]                          | —    | —  | ✓     | —    | —  | —             |
| <i>Language-quality evaluation</i>      |      |    |       |      |    |               |
| MQM [11]                                | —    | —  | ✓     | —    | —  | ✓             |
| MENLO [12]                              | —    | —  | ✓     | —    | —  | ✓             |
| <b><math>\tau</math>-Multilingual</b>   | ✓    | ✓  | ✓     | ✓    | ✓  | ✓             |

executed tool tasks [6]; VoiceAgentBench evaluates multilingual spoken tool use at the query/workflow level [7]; and Speech-MASSIVE evaluates intent and slot prediction from recorded speech [8]. X-RiSAWOZ and GlobalWoZ provide multilingual text dialogue with human-verified or locale-native content [9, 10]; MQM and MENLO motivate factorized, human-reviewed language evaluation [11, 12]. Table 1 shows that no prior benchmark jointly evaluates multilingual full-duplex speech, executable task success, interaction quality, and language-quality auditing;  $\tau$ -Multilingual combines these capabilities with native-speaker-authored diagnostic factors.

### 3. METHODOLOGY

#### 3.1. Matched Multilingual Benchmark

$\tau$ -Voice supplies English task frames, agent policies and tools, user scenarios, and a simulated caller [2]. Its full-duplex runtime advances in 200-ms ticks; binary rewards check task-specified environment, action, and communication criteria. Each language pack localizes the caller instructions, persona, voice, identity entities, and evaluation rules while preserving the underlying tasks and scoring. The same specification drives both voice and matched text conditions, enabling comparisons by domain, task, system, and reasoning setting. LanguageFactory stores this specification as a typed, versioned artifact; Table 2 illustrates the Hindi pack. Cross-language differences therefore compare complete localized systems rather than an isolated causal effect of language.

**Table 2.** LanguageFactory components illustrated with the Hindi pack. Entity localization maps the same source identity to locale-specific backend values without changing the task.

| Language component                | Hindi (India)                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------|
| Personas                          | Rishika uses fast Hinglish; Imran uses patient, Urdu-inflected Hindi.                |
| Speech rules                      | Spell with “R for Rajdhani”; say <i>at</i> and <i>dot</i> in email addresses.        |
| Task arms and entity localization | Allison Reeves $\rightarrow$ Neha Gupta; phone: 99400 97181.                         |
| Language-quality factors          | Customer address ( <i>aap</i> vs. <i>tum/tu</i> ) and agreement with <i>aap</i> .    |
| Review and freeze                 | Pin the Rishika and Imran voices; freeze preset <code>multilingual.v1.hindi</code> . |

Native speakers review each pack’s language rules, voices, and representative end-to-end calls. Automated checks enforce schema, task parity, and persona assignment before release.

#### 3.2. Evaluation Cohorts

The benchmark contains 900 localized task instances: 50 tasks in each of three domains, instantiated in six languages. Five voice configurations yield 4,500 primary calls: GPT (OpenAI `gpt-realtime-2`) at minimal and xhigh reasoning, Gemini 3.1 Flash Live Preview at minimal and high, and Grok Voice Think Fast 1.0 [15, 16, 17]. The cascaded caller uses `gpt-5.5 xhigh` for text and Eleven v3 for speech [18, 19]. For task-completion significance tests only, the four GPT and Gemini configurations include one additional trial. We use the short labels GPT, Gemini, and Grok below.

The matched half-duplex text control evaluates the same 900 localized task instances with two systems (1,800 calls): GPT-5.5 xhigh and Gemini 3.1 Pro Preview high [18, 20]. Because both the model and interaction regime change, this is an end-to-end control rather than a pure acoustic intervention.

#### 3.3. Retail Localization Ablations

We evaluate GPT xhigh and Gemini high on the same 30 Hindi and Mandarin retail tasks. The source-English identity arm restores source-English identities while preserving the target-language conversation. The native-script database arm replaces romanized localized names with native-script names and matched native spell-out. Each cell has one trial per task and is reported descriptively.

#### 3.4. Evaluation Measures

Task completion is the call-level environment-reward pass rate. *Interaction* equals 100 minus the mean of five call-level failure rates: non-response (late/missed replies or failure to yield), interruption (talk-over), selectivity (responses to backchannels, vocal tics, or non-directed speech), monologues over 45 seconds, and validated tool misuse (incorrect arguments or agent-caused errors). *Generation quality* (Generation) is the share of aligned agent utterances passing naturalness and speech-fidelity checks. Naturalness flags unnatural phrasing; fidelity uses a multimodal Gemini 3.1 Pro Preview judge [20] to compare audio with expected text, counting severity- $\geq 2$  findings or Mandarin meaning-changing tone errors as failures. These composites are uncalibrated summaries; language-specific diagnostics remain separate. English is excluded only from Generation because our Naturalness rubric targets translationese and native phrasing in the five localized languages.

We compare systems within languages and estimate matched performance gaps relative to English for Task completion and Interaction. We do not rank languages: each is a complete localized system, and naturalness is not cross-language calibrated.

#### 3.5. Human Validation

LLM-based judges enter results only when precision and recall exceed .75, F1 exceeds .80, and Cohen’s  $\kappa > .60$ . Naturalness and language-specific diagnostics require consensus from two native-language annotators. Validation covers tool use, about 60 examples per language for naturalness and speech fidelity, and at least 30 per diagnostic; failing measures remain archived.

Separately, reviewers audit 30 completed calls per language, balanced across domains and systems. They attribute the first critical error to the agent, simulator, or infrastructure and score eight caller-quality dimensions—including voice/prosody quality, turn-taking naturalness, and behavioral plausibility—on a four-point scale (see the released audit rubric [5]).

#### 3.6. Statistical Analysis

Task-completion inference averages two trials per task for the four GPT/Gemini configurations and compares each language with English over 150 matched domain–task clusters using paired permutation tests over five contrasts and paired BCa 95% intervals from 10,000 resamples. Grok’s single trial appears only in Figure 2(a)’s descriptive estimates.

Interaction and Generation use 100,000 paired cluster-level label swaps. Interaction swaps the five systems and reagggregates

|           | (a) Task completion |    |    |    |    |    | (b) Interaction |    |    |    |    |    | (c) Generation |    |    |    |    |
|-----------|---------------------|----|----|----|----|----|-----------------|----|----|----|----|----|----------------|----|----|----|----|
|           | EN                  | ES | PT | HI | KO | ZH | EN              | ES | PT | HI | KO | ZH | ES             | PT | HI | KO | ZH |
| GPT min.  | 45                  | 46 | 49 | 43 | 26 | 37 | 52              | 46 | 46 | 48 | 37 | 40 | 64             | 69 | 58 | 61 | 73 |
| GPT xhigh | 65                  | 58 | 69 | 61 | 33 | 47 | 47              | 40 | 43 | 45 | 32 | 39 | 68             | 74 | 62 | 62 | 71 |
| Gem min.  | 41                  | 37 | 38 | 39 | 29 | 31 | 49              | 44 | 43 | 48 | 36 | 35 | 58             | 67 | 47 | 66 | 59 |
| Gem high  | 51                  | 60 | 59 | 60 | 44 | 51 | 45              | 42 | 45 | 42 | 35 | 36 | 61             | 71 | 49 | 57 | 52 |
| Grok      | 76                  | 78 | 79 | 76 | 72 | 70 | 50              | 47 | 47 | 44 | 39 | 34 | 50             | 60 | 40 | 49 | 63 |
| Overall   | 56                  | 56 | 59 | 56 | 41 | 47 | 49              | 44 | 45 | 45 | 36 | 37 | 60             | 68 | 51 | 59 | 64 |

**Fig. 2.** Headline outcomes by system and language. Higher is better; color scales normalize within panel, outlines mark language-best systems, Overall averages the five configurations, and Generation excludes English.

| Metric                                   | GPT min.    | GPT xhigh | Gem min.  | Gem high  | Grok      |
|------------------------------------------|-------------|-----------|-----------|-----------|-----------|
| Task failure <sub>c</sub> (%)            | 59          | 44        | 64        | 46        | <b>25</b> |
| Non-response <sub>c</sub> (%)            | <b>8</b>    | 28        | 48        | 77        | 10        |
| Interruption <sub>c</sub> (%)            | 81          | 90        | <b>75</b> | 80        | 89        |
| Selectivity error <sub>c</sub> (%)       | 97          | 96        | 83        | <b>77</b> | 97        |
| Monologue <sub>c</sub> (%)               | 19          | 19        | <b>6</b>  | 10        | 34        |
| Tool misuse <sub>c</sub> (%)             | 69          | 62        | 76        | 53        | <b>51</b> |
| Naturalness failure <sub>u</sub> (%)     | 32          | <b>30</b> | 37        | 38        | 43        |
| Speech-fidelity failure <sub>u</sub> (%) | 8           | <b>7</b>  | 8         | 9         | 21        |
| Turn-taking latency <sub>t</sub> (s)     | <b>1.12</b> | 1.42      | 1.36      | 1.82      | 1.20      |

**Fig. 3.** Main-metric breakdown across languages by system; lower is better. Bold numbers mark row minima. Subscripts c/u/t denote call/utterance/turn; units are in parentheses.

component failure/opportunity counts; Generation swaps the five localized calls and reaggregates aligned utterances. Holm correction covers five language contrasts for Task completion and Interaction and ten system pairs for Generation. The retail ablations are descriptive.

## 4. RESULTS

Figures 2 and 3 summarize task completion, Interaction, Generation, and their component measures. We then examine localization failures and model choice.

### 4.1. Task Completion

Task completion is broadly preserved in Spanish, Portuguese, and Hindi but not in Korean and Mandarin (Figure 2(a)). The confirmatory English-relative losses for Korean and Mandarin are 18.8 points (95% CI: 14.8–23.3) and 9.5 points (6.0–13.4), respectively; both are Holm-adjusted  $p < .00005$ , while the other three differences are not significant. The domain pattern is similarly misaligned: telecom is hardest by task completion (44.1%) yet yields the highest Generation (63.2%).

The voice-text gap favors pooled text systems in every language by 16.8–41.1 points. The smallest gap is Spanish; the largest is Korean, where text reaches 82.0% versus 40.9% voice. Because model and turn-taking also differ, this is an end-to-end

voice-system gap rather than a purely acoustic effect.

### 4.2. Interaction

All English-relative Interaction declines are significant: 4.6, 3.7, 3.1, 12.6, and 11.8 points for Spanish, Portuguese, Hindi, Korean, and Mandarin (Holm-adjusted  $p \leq .00055$ ).

Korean combines non-response (46.3% versus 27.8% in English) and tool-use failure (84.8% versus 49.8%); Mandarin combines interruption (90.9% versus 78.0%) and tool-use failure (77.4%). Selectivity failure is high throughout (86.7–93.5%), indicating a general weakness rather than the main language-gap driver. Spanish and Portuguese show more monologuing than English but not the Korean/Mandarin pattern, so similar aggregate losses require different remedies.

### 4.3. Generation Quality

Generation produces a different model ranking: GPT xhigh leads Spanish, Portuguese, and Hindi; Gemini minimal leads Korean; and GPT minimal leads Mandarin (Figure 2(c)).

In their naturalness and cultural-grounding audits, native reviewers found recurring literal or interface-like language: Spanish *user id*, Portuguese “nenhum voo ainda foi voador” (no flight has yet been flown), Hindi “explicit yes,” Korean spoken tool names such as *check\_sim\_status*, and Mandarin *toggle* data. Targeted audits capture different conventions: mixed formal and informal address in Portuguese; gendered self-reference and omitted *-ji* in Hindi; missing subject honorification in Korean; and unsuitable titles and missing sentence-final particles in Mandarin. These diagnostics contextualize Naturalness but are not components of Generation. GPT is more natural in Hindi than Gemini and Grok (35.1–40.6% versus 49.7–55.5% failure), yet GPT minimal has the most gender errors (49.3%). In Mandarin, GPT has nearly all modal-particle failures (40.0–61.1% versus 0–0.7%).

A related, separate diagnostic is gender agreement. Agent gender was inadvertently absent from the prompt despite using female voices, yielding inconsistent self-grounding. Hindi agents default to masculine self-reference (49.5% of calls versus 5.3% with any feminine form). The overall gender-agreement failure rates—27.2% in Hindi and 29.6% in Portuguese—combine self-reference, noun/anaphora agreement, and caller-directed errors. Disclosure changes the mix but does not reliably reduce the total:

in 30-call-per-language ablations, failures rise from 30.0% to 40.0% in Hindi and fall from 30.7% to 20.0% in Portuguese.

Speech fidelity reveals an all-language provider pattern: 80.0% of Grok utterances contain no material audible error, versus 91.3–93.8% for the other systems. Across 72,164 scored utterances, the most common material audible errors are audible punctuation or text-formatting artifacts (3,467 utterances), mispronunciations (1,389), and number, date, or currency errors (1,094).

#### 4.4. Localization and Entity Handling

The large Korean and Mandarin task losses concentrate in authentication: exact lookup succeeds in 81–85% of English calls, but only 26–56% of Korean and 60–74% of Mandarin calls across repeated GPT/Gemini systems. Localized caller prompts provide the correct family-first name, yet agents often map the spoken parts into a given-first lookup. In Mandarin, this can break exact lookup even when callers provide the correct romanized spelling and the agent prompt requires all tool calls, arguments, and structured outputs in English; native-character explanations add a separate cross-script mismatch. Telecom’s balanced split shows an exploratory crossover: Korean task completion is 32.0% for name+DOB versus 35.2% for phone authentication, whereas Mandarin is 40.8% versus 33.6%; method remains bundled with task.

**Table 3.** Retail entity ablations: task completion (%) on 30 tasks.

| Lang.    | System      | Entity    |        | Entity + DB |        |
|----------|-------------|-----------|--------|-------------|--------|
|          |             | Localized | Source | Romanized   | Native |
| Hindi    | GPT xhigh   | 63.3      | 63.3   | 63.3        | 36.7   |
|          | Gemini high | 60.0      | 66.7   | 60.0        | 60.0   |
| Mandarin | GPT xhigh   | 36.7      | 20.0   | 36.7        | 53.3   |
|          | Gemini high | 53.3      | 53.3   | 53.3        | 53.3   |

In the localization ablations, source-English identities match or outperform the localized baseline in both Hindi cells and Mandarin Gemini, but lower Mandarin GPT by 16.7 points. Replacing romanized database values with native-script names improves Mandarin GPT by 16.7 points and leaves Mandarin Gemini unchanged. Hindi differs: native script leaves Gemini unchanged and lowers GPT by 26.7 points. These 30-task cells are exploratory; exact authentication does not ensure task completion.

#### 4.5. Model Choice Across Outcomes

Figure 3 decomposes the model-level tradeoffs. No single configuration leads every headline outcome; model choice should follow the deployment objective:

- **Task completion:** Grok leads task completion in every language, with a six-language mean of 75.1%, but has the lowest Generation score (52.3%).
- **Quality:** GPT xhigh leads Generation (67.5%) at 55.7% task completion; GPT minimal leads Interaction (44.9%) and has the lowest turn-taking latency (1.12 s).
- **Stability:** The narrowest descriptive cross-language ranges are Grok’s 8.7 points in task completion and Gemini high’s 10.1

points in Interaction. We do not interpret Generation ranges because Naturalness is not calibrated across languages.

- **Latency:** Higher-reasoning configurations show longer turn-taking latency (GPT: 1.12–1.42 s; Gemini: 1.36–1.82 s). Mean call durations are 11.8–14.2 min for GPT/Gemini versus 24.7 min for Grok.

#### 4.6. Validation and Robustness

Pooled held-out F1 is 96% for tool use and 88% for both naturalness and speech fidelity; the remaining Interaction components are deterministic. Among measures meeting the validation thresholds, naturalness spans Hindi (F1 81.4%,  $\kappa = .634$ ) to Spanish (F1 97.9%,  $\kappa = .965$ ), and speech fidelity spans Korean (F1 82.4%,  $\kappa = .754$ ) to Spanish (F1 96.6%,  $\kappa = .955$ ).

The separate caller audit also supports the evaluation contract. Reviewers attribute only 5/180 first critical errors (2.78%) and 5/104 failed calls (4.81%) to the simulator, and none to infrastructure. The eight caller-quality dimensions average 3.05/4; backchannel naturalness is lowest at 2.86.

All system pairs differ significantly on utterance Generation; the weakest is Gemini minimal versus high (1.9 points; Holm-adjusted  $p = .02378$ ). Interaction effects preserve modest Spanish/Portuguese/Hindi versus large Korean/Mandarin degradation.

### 5. LIMITATIONS

We study six languages, three domains, three hosted model families, and one speech stack. No documented open-weight model covered all five localized languages, full-duplex speech, and live tool use; cost limited additional languages, models, and trials. Simulated callers may not reproduce human timing or repair. The caller consumes agent transcripts to minimize simulator-side ASR noise, so synthesis errors cannot affect task outcomes. GPT and Gemini have two trials versus Grok’s one, and validation sets are finite. The frozen validation cohort includes earlier Korean and Mandarin retail runs whose caller prompts underspecified first- and last-name roles.

### 6. CONCLUSION

$\tau$ -Multilingual shows that multilingual voice-agent reliability extends beyond task completion. Spanish, Portuguese, and Hindi largely match English, whereas Korean and Mandarin lose ground through authentication, entity, and interaction failures. Because model rankings depend on the objective, jointly reporting task completion, interaction, and generation exposes failures hidden by English-only or outcome-only benchmarks. We release language packs, validated judges, and tools for community-built evaluation.

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## 7. COMPLIANCE WITH ETHICAL STANDARDS

The benchmark uses no real customer records or recordings. Records are synthetic and non-sensitive; calls use simulated callers and provider voices. Reviewers authored benchmark materials and annotated outputs; they consented and were compensated. Before release, artifacts were screened for incidental personal data or secrets, and flagged items were excluded.

The benchmark evaluates voice-agent systems, not people or languages; its results should not be interpreted as measures of human ability or cultural quality.

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